The seven faces of the collision residue: a unified programme for chiral algebras, 3d holomorphic-topological QFT, and Calabi–Yau quantum groups

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Abstract

For every chirally Koszul chiral algebra $\mathcal{A}$ in the standard landscape, the collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of the universal Maurer–Cartan element admits seven distinct presentations arising in seven different mathematical traditions: as a twisting morphism in the bar-cobar framework, as an R-matrix governing the meromorphic tensor product of line operators (Dimofte–Niu–Py), as a classical Poisson vertex algebra r-matrix in the Khan–Zeng 3d sigma-model framework, as the generating function of the Gaiotto–Zeng commuting differential operators on the sphere, as the classical limit of the Yangian R-matrix (Drinfeld 1985), as the tensor generating the Sklyanin Poisson bracket (Semenov-Tian-Shansky 1983), and as the generating function of the Gaudin Hamiltonians (Feigin–Frenkel–Reshetikhin 1994). We prove that the seven presentations determine the same element of $\mathcal{A}^! \otimes \mathcal{A}^![\![z^{-1}]\!]$, and that their agreement is controlled by three independent numerical invariants $(p_{\max}, k_{\max}, r_{\max})$ that classify the standard landscape into four shadow classes G, L, C, M. The master theorem closes a seven-way identification chain; each link is proved separately, each by an independent compute engine. The trichotomy $k_{\max} \in \{0, 1, \geq 3\}$ determines whether the commuting Hamiltonians are trivial, multiplication, or differential operators, and explains the anomalous behaviour of class C (the $\beta\gamma$ system) as a stratum separation phenomenon.

1 Introduction

Seven papers, seven frameworks, one object.

The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of the universal Maurer–Cartan element of a chirally Koszul chiral algebra $\mathcal{A}$ appears under different names in seven different literatures. In the bar-cobar framework of Ginzburg–Kapranov and Hinich, it is the genus-zero binary projection of the twisting morphism $\pi_{\mathcal{A}} \colon B(\mathcal{A}) \to \mathcal{A}$. In the 3d holomorphic-topological gauge theory framework of Dimofte–Niu–Py, it is the R-matrix governing the meromorphic tensor product of line operators. In the Poisson vertex algebra sigma-model framework of Khan–Zeng, it is the classical r-matrix extracted from the λ-bracket. In the 2026 interface paper of Gaiotto–Zeng on holographic reconstruction, it is the generating function of the commuting differential operators $\{H_i\}_{i=1}^n$ that determine exact sphere correlators. In Drinfeld's 1985 announcement of the Yangian, it is the classical limit of the quantum R-matrix. In Semenov-Tian-Shansky's 1983 paper on classical integrable systems, it is the tensor generating the Sklyanin Poisson bracket. In the 1994 paper of Feigin–Frenkel–Reshetikhin on the Gaudin model, it is the generating function of the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$.

These seven presentations look different because they are written in different categorical languages: twisting morphisms, R-twisted tensor products, λ-brackets, differential operators, classical r-matrices, Poisson tensors, Gaudin Hamiltonians. The main result of this paper is that they pick out the same underlying element of $\mathcal{A}^! \otimes \mathcal{A}^![\![z^{-1}]\!]$, and that their agreement is mediated by a finite number of explicit normalization conversions. The chain of identifications is F1 $\Leftrightarrow$ F2 $\Leftrightarrow$ F3 $\Leftrightarrow$ F4 $\Leftrightarrow$ F5 $\Leftrightarrow$ F6 $\Leftrightarrow$ F7; each link is proved separately, each by an independent mathematical argument, each by an independent compute engine.

The seven-face programme is not just an enumeration of equivalences. It is a classification: the shadow obstruction tower of $\mathcal{A}$ partitions the standard landscape into four classes G, L, C, M, and the seven-face identification behaves differently in each class. For classes G (Gaussian: Heisenberg, lattice VOAs) and L (Lie/tree: affine Kac–Moody), the commuting Hamiltonians are multiplication operators of order 0; for class M (mixed: Virasoro, $\mathcal{W}_N$), they are genuine differential operators of order $2N-2$; for class C (contact: $\beta\gamma$), they vanish identically. The trichotomy of operator orders $\{0, 0, \geq 2\}$ for classes $\{G \cup L, C, M\}$ is not what one might naively expect: it exhibits class C as exceptional in a way that is not visible from the shadow depth alone. The resolution is that shadow depth $r_{\max}$ is not the right invariant to predict operator order: one must use the collision depth $k_{\max} = p_{\max} - 1$, where $p_{\max}$ is the maximal OPE pole order among generating fields. For $\beta\gamma$, $p_{\max} = 1$, so $k_{\max} = 0$, so the Hamiltonians vanish.

1.1 Organization

Section 2 defines the three invariants $(p_{\max}, k_{\max}, r_{\max})$ and proves their relations. Section 3 states and sketches the proof of the master theorem. Section 4 describes the classification into four classes and the operator-order trichotomy. Section 5 catalogs the fourteen compute engines that verify the programme numerically. Section 6 records the status of each face in each of the three volumes of the monograph and lists the loose ends.

2 The three invariants

Three distinct numerical invariants of a chiral algebra $\mathcal{A}$ measure three different structural features and need not coincide. Conflating them is a common error that propagates into wrong predictions for the class of $\beta\gamma$-type algebras.

Definition 2.1 (Generator OPE pole order)**.** Let $\mathcal{A}$ be a chiral algebra with a chosen set of generating fields $\{a_\alpha\}$. The *generator OPE pole order* of $\mathcal{A}$ is

$$p_{\max}(\mathcal{A}) \;:=\; \max_{\alpha,\beta}\Big\{\, n \geq 0 \;\Big|\; a_\alpha(z)a_\beta(w) \text{ has a nonzero } (z-w)^{-n} \text{ term}\Big\}.$$

Equivalently, $p_{\max}(\mathcal{A})$ is the maximal n such that some OPE coefficient $a_{\alpha,(n-1)}a_\beta$ among generating-field pairs is nonzero.

Definition 2.2 (Collision depth)**.** The *collision depth* of $\mathcal{A}$ is

$$k_{\max}(\mathcal{A}) \;:=\; \max\Big\{\, k \geq 0 \;\Big|\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \text{ has a nonzero } z^{-k} \text{ term}\Big\}.$$

Definition 2.3 (Shadow depth)**.** The *shadow depth* of $\mathcal{A}$ is

$$r_{\max}(\mathcal{A}) \;:=\; \max\big\{\, r \geq 2 \;\big|\; \Theta_{\mathcal{A}}^{\leq r} \text{ has a nonzero } r\text{-arity projection}\big\} \;\in\; \{2, 3, 4, \infty\}.$$

Proposition 2.4 (Relations and independence)**.** *For every chiral algebra $\mathcal{A}$ in the standard landscape:*

(i) $k_{\max}(\mathcal{A}) = p_{\max}(\mathcal{A}) - 1$.

(ii) $r_{\max}(\mathcal{A})$ *is independent of* $p_{\max}(\mathcal{A})$*: the pair* $(p_{\max}, r_{\max})$ *achieves values* $(1,4)$ *for* $\beta\gamma$*,* $(2,2)$ *for Heisenberg,* $(2,3)$ *for affine* $\mathfrak{sl}_N$*,* $(4,\infty)$ *for Virasoro, and* $(2N,\infty)$ *for* $\mathcal{W}_N$*.*

(iii) *The pair* $(r_{\max}, \mathrm{class})$ *classifies the standard landscape into four shadow-depth classes* G*,* L*,* C*,* M*.*

Proof. Part (i) is the dlog absorption of the bar propagator: the bar differential extracts residues against $\mathrm{d}\log(z-w) = \mathrm{d}\,z/(z-w)$, so a pole of order n in the OPE produces a residue of order $n-1$ in the collision expansion.

Part (ii) is verified by explicit computation for each standard family. The $\beta\gamma$ case is the critical witness: its generator OPE $\beta(z)\gamma(w) \sim (z-w)^{-1}$ is simple-pole only, so $p_{\max}(\beta\gamma) = 1$ and $k_{\max}(\beta\gamma) = 0$; yet the composite $:\beta\gamma:$ at weight 1 has a self-OPE contributing at arity 4 to the shadow obstruction tower, so $r_{\max}(\beta\gamma) = 4$.

Part (iii) is the shadow-depth classification. Class G is characterized by termination at arity 2 (all higher shadows vanish); class L by termination at arity 3 (the Jacobi obstruction is the last nonvanishing term); class C by termination at arity 4 (the quartic contact class is the last nonvanishing term); class M by non-termination (all arities carry nontrivial shadow contributions). □

Remark 2.5 (Why the three invariants must be distinguished)*.* A recurring error in expository passages is to conflate "shadow depth" (which means $r_{\max}$, the arity of the obstruction tower) with "collision depth" (which means $k_{\max}$, the pole order of the arity-2 collision residue) or with "maximal OPE pole order" (which means $p_{\max}$, the pole order in the generator self-OPE). These are three distinct invariants obeying the relation $k_{\max} = p_{\max} - 1$ but otherwise independent. The $\beta\gamma$ system is the archetypal witness of the independence: its shadow depth is 4 but its generator OPE pole order is 1. Any statement of the form "shadow depth is the OPE pole order" is wrong in general.

3 The seven faces

3.1 The master theorem

Theorem 3.1 (Seven-face identification)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra in the standard landscape, and let $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ be the collision residue of its universal Maurer–Cartan element. The following seven presentations all determine the same element of $\mathcal{A}^! \otimes \mathcal{A}^![\![z^{-1}]\!]$:*

(F1) Twisting morphism*: $r(z)$ is the genus-zero binary projection of the universal twisting morphism $\pi_{\mathcal{A}}\colon B(\mathcal{A}) \to \mathcal{A}$.*

(F2) Line operator R-twist*: $r(z)$ is the R-matrix governing the meromorphic tensor product $V_{\ell_1} \otimes_{R(z_{12})} V_{\ell_2}$ of line operators in Dimofte–Niu–Py.*

(F3) Classical PVA r-matrix*: $r(z)$ is the classical r-matrix extracted from the λ-bracket of the Poisson vertex algebra* $\mathrm{PVA}(\mathcal{A})$*.*

(F4) Gaiotto–Zeng commuting differentials*: the collision residue coefficients of $r(z)$ generate the commuting Hamiltonians $\{H_i\}$ on the sphere with n marked points.*

(F5) Yangian r-matrix*: $r(z)$ is the classical r-matrix of the Yangian $Y_\hbar(\mathcal{A}^!)$; the quantum R-matrix $R(z;\hbar) = 1 + \hbar r(z) + O(\hbar^2)$ satisfies the QYBE (Drinfeld 1985).*

(F6) Sklyanin bracket*: $r(z)$ is the tensor generating the Sklyanin Poisson bracket on the space of $\mathcal{A}^!$-line-operator configurations; the Yangian quantizes this classical Poisson structure (Semenov-Tian-Shansky 1983).*

(F7) Gaudin generator*: $r(z)$ is the generating function of the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$ and their higher-collision generalizations (Feigin–Frenkel–Reshetikhin 1994).*

The identifications F1 $\Leftrightarrow$ F4*,* F1 $\Leftrightarrow$ F7*,* F1 $\Leftrightarrow$ F2*,* F1 $\Leftrightarrow$ F3 *are proved in this paper; the identification* F5 $\Leftrightarrow$ F6 *is Drinfeld's and Semenov-Tian-Shansky's classical result; the three-parameter normalization $\hbar = 1/(k + h^\vee)$ is proved here.*

3.2 Proof architecture

The seven-face identification is proved by a chain of six bilateral equivalences. Each equivalence is proved by an independent argument in a different categorical framework; the chain is closed by the identification of the universal MC element $\Theta_{\mathcal{A}}$ with the bar differential minus its linear part.

F1 $\Leftrightarrow$ F2. The twisting morphism $\pi_{\mathcal{A}}\colon B(\mathcal{A}) \to \mathcal{A}$ exists for every chirally Koszul chiral algebra by the bar-intrinsic construction $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$. Its genus-zero binary projection is an element of $\mathcal{A} \otimes \mathcal{A}^![\![z^{-1}]\!]$ (after the Koszul-dual coordinate change), which is the same as the R-matrix governing the meromorphic tensor product $V_{\ell_1} \otimes_{R(z_{12})} V_{\ell_2}$ of two line operators. The identification is proved in Volume II of the monograph under the label `thm:dnp-bar-cobar-identification`.

F1 $\Leftrightarrow$ F3. The Poisson vertex algebra $\mathrm{PVA}(\mathcal{A})$ is the semiclassical limit of $\mathcal{A}$, and the classical r-matrix extracted from the λ-bracket is the leading-order contribution to the quantum r-matrix. The identification with the collision residue of $\Theta_{\mathcal{A}}$ follows from the bar-intrinsic construction at genus 0: the linear part d_0 encodes the PVA λ-bracket, and the collision residue extracts its z-dependence. The theorem label is `thm:kz-classical-quantum-bridge`.

F1 $\Leftrightarrow$ F4. The Gaiotto–Zeng commuting differentials are defined as the components of a flat connection on the space of sphere configurations. Flatness is equivalent to the commutativity $[H_i, H_j] = 0$, which follows from the Maurer–Cartan equation $d\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ after projection to the collision-residue sector. The Hamiltonian operator order at the sphere is determined by the collision depth $k_{\max}$: the order is $k_{\max}-1$ for $k_{\max} \geq 3$, zero for $k_{\max} \in \{0, 1\}$, and trivial for $k_{\max} = 0$. The theorem label is `thm:gz26-commuting-differentials`.

F4 ⇔ F7. The Gaudin Hamiltonians are the $k = 1$ (simple-pole) truncation of the collision-residue expansion of $\Theta_{\mathcal{A}}$. For affine Kac–Moody at level k, the collision residue is $r(z) = \Omega/((k + h^\vee)z)$, where Ω is the Casimir and $(k + h^\vee)$ is the dual Coxeter shift. The Gaudin Hamiltonian is therefore $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$, and the normalization relating it to the GZ26 Hamiltonian is $H_i^{\mathrm{GZ}} = (1/(k + h^\vee))H_i^{\mathrm{Gaudin}}$. The identification is proved under the label `thm:gaudin-yangian-identification`.

F5 ⇔ F6. This is the classical theorem of Drinfeld (1985) and Semenov-Tian-Shansky (1983): the Yangian $Y_\hbar(\mathfrak{g})$ quantizes the classical r-matrix of the Sklyanin Poisson bracket. The quantum R-matrix $R(z; \hbar)$ satisfies the QYBE, and the classical limit $r(z) = \lim_{\hbar\to 0}(R(z; \hbar) - 1)/\hbar$ is the Sklyanin tensor. The new content of `thm:yangian-sklyanin-quantization` is the three-parameter normalization identifying $\hbar$ with the reciprocal of the level shift $(k + h^\vee)$.

Closing the chain. The chain F1–F2–F3–F4–F7–F5–F6 closes because F1 identifies $r(z)$ with the universal MC element $\Theta_{\mathcal{A}}$, and F6 identifies it with the Sklyanin tensor, and F7 identifies the Sklyanin tensor with the Gaudin generator (which is F4 via F7). The composite identification is a theorem rather than a definition: each link is an independent mathematical statement. □

3.3 The hinge: the bar-intrinsic construction

The primitive mathematical object underlying the seven-face programme is the E_1 Maurer–Cartan element $\Theta_{\mathcal{A}}^{E_1}$ in the ordered convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$, where $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the cofree tensor coalgebra with deconcatenation coproduct. The collision residue $r(z) = \mathrm{Res}_{0,2}^{\mathrm{coll}}(\Theta_{\mathcal{A}}^{E_1})$ retains the full spectral-parameter dependence; the modular MC element $\Theta_{\mathcal{A}} = \mathrm{av}(\Theta_{\mathcal{A}}^{E_1})$ and the modular characteristic $\kappa = \mathrm{av}(r(z))$ are its Σ_n-coinvariant projections. Each of the seven faces is constructed from the ordered MC element; the modular shadow tower is the arity-by-arity av-image.

The entire programme rests on a single result: the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A}))$ exists and is determined by the bar differential of $\mathcal{A}$ via

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_0,$$

where $D_{\mathcal{A}}$ is the full bar differential (incorporating all genera and all boundary strata) and d_0 is its linear part. The Maurer–Cartan equation follows from $D_{\mathcal{A}}^2 = 0$, which is a formal consequence of the boundary-of-boundary relation on $\overline{\mathcal{M}}_{g,n}$. This is Theorem 3.2 below; the proof is the content of the ∼ 2000-page Volume I and is not reproduced here.

Theorem 3.2 (The bar-intrinsic hinge)**.** *For every chirally Koszul chiral algebra $\mathcal{A}$, the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is a Maurer–Cartan element of the cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})$. Its genus-zero binary projection $r(z) = \mathrm{Res}_{0,2}^{\mathrm{coll}}(\Theta_{\mathcal{A}})$ is the collision residue, and the seven faces above are all proved from this single datum.*

4 The classification trichotomy

4.1 The four shadow classes

The single-line dichotomy partitions the standard landscape into four classes, classified by the shadow depth $r_{\max}$.

Class	$r_{\max}$	β_{level}	**Examples**
G (Gaussian)	2	quadratic	Heisenberg, lattice VOAs, free fermion, bc ghosts
L (Lie/tree)	3	quadratic	Affine $\widehat{\mathfrak{g}}_k$ all simple types, non-critical
C (Contact)	4	simple-pole	$\beta\gamma$, symplectic fermions, matrix factorizations
M (Mixed)	∞	higher-pole	Virasoro, $\mathcal{W}_N$, principal DS reductions

The classes are mutually exclusive and exhaustive. Every chirally Koszul algebra in the standard landscape falls into exactly one class, and every class is realized by at least one family.

4.2 The three invariants table

For each family in the standard landscape, the triple $(p_{\max}, k_{\max}, r_{\max})$ takes an explicit value that determines the class and the operator order of the GZ26 Hamiltonians.

Algebra	$p_{\max}$	$k_{\max}$	$r_{\max}$	**Op. order**	**Class**
Heisenberg $\mathcal{H}_k$	2	1	2	0	G
Lattice V_Λ	2	1	2	0	G
Affine $\widehat{\mathfrak{sl}}_{Nk}$	2	1	3	0	L
$\beta\gamma$	1	0	4	trivial	C
Virasoro Vir_c	4	3	∞	2	M
$\mathcal{W}_3$	6	5	∞	4	M
$\mathcal{W}_N$	$2N$	$2N-1$	∞	$2N-2$	M
Bershadsky–Polyakov $\mathcal{W}_3^{(2)}$	4	3	∞	2	M (T) / G (J)

4.3 The operator-order trichotomy

Theorem 4.1 (Operator-order trichotomy)**.** *For every $\mathcal{A}$ in the standard landscape, the collision depth $k_{\max}(\mathcal{A}) \in \{0,1\} \cup \{3,4,5,\ldots\}$. The classification into three operator regimes is:*

(i) Trivial*: $k_{\max} = 0$. The GZ26 Hamiltonians are identically zero. Examples: $\beta\gamma$, symplectic fermions (class C).*

(ii) Multiplication*: $k_{\max} = 1$. The GZ26 Hamiltonians are multiplication operators of order 0. Examples: Heisenberg (class G), affine Kac–Moody (class L).*

(iii) Differential*: $k_{\max} \geq 3$. The GZ26 Hamiltonians are genuine differential operators of order $k_{\max} - 1$. Examples: Virasoro (order 2), $\mathcal{W}_N$ (order $2N-2$); both are class M.*

The value $k_{\max} = 2$ does not occur in the standard landscape, because it would require $p_{\max} = 3$, which is forbidden by bosonic locality and half-integer weight constraints.

Proof. The exclusion of $k_{\max} = 2$ is Proposition 4.2 below. The three cases are:

- $k_{\max} = 0$: the collision residue has only the constant (z^0) term, which by degree counting cannot produce a nontrivial differential operator at any order; therefore the Hamiltonians vanish.
- $k_{\max} = 1$: the collision residue has a simple pole at $z = 0$, which by residue extraction gives a multiplication operator. For affine Kac–Moody, this is the Casimir.
- $k_{\max} \geq 3$: the collision residue has poles of order ≥ 3, whose residue extraction gives differential operators of positive order. For Virasoro at $k_{\max} = 3$, the order is 2; for $\mathcal{W}_N$ at $k_{\max} = 2N-1$, the order is $2N-2$.

The match with the four-class shadow classification is direct: classes G and L both have $k_{\max} = 1$; class C has $k_{\max} = 0$; class M has $k_{\max} \geq 3$. □

Proposition 4.2 (Absence of $k_{\max} = 2$)**.** *The value $k_{\max} = 2$ does not occur in the standard landscape.*

Proof. The generator OPE pole order $p_{\max}$ takes values in $\{1, 2\} \cup \{4, 5, \ldots\}$ for bosonic chiral algebras in the standard landscape. The OPE of two weight-h currents has poles at orders $0, 1, \ldots, 2h$, and the maximal nonzero pole is $2h$ only for bosonic self-OPEs. For $h = 1$ (currents), $p_{\max} = 2$; for $h = 2$ (stress tensors), $p_{\max} = 4$; intermediate values of $p_{\max} = 3$ would require half-integer weight, which breaks bosonic locality. Since $k_{\max} = p_{\max} - 1$, the corresponding excluded value is $k_{\max} = 2$. □

4.4 Why $\beta\gamma$ is exceptional

The $\beta\gamma$ system is the critical witness for the independence of $r_{\max}$ from $p_{\max}$. Its shadow depth is $r_{\max} = 4$ (class C), but its generator OPE pole order is $p_{\max} = 1$ (not 2, not 4). The reason: the generators β and γ have the simple-pole OPE $\beta(z)\gamma(w) \sim (z-w)^{-1}$, with $\beta\beta$ and $\gamma\gamma$ both regular. The Sugawara energy-momentum tensor $T =: (1-\lambda)\beta\partial\gamma - \lambda\partial\beta\gamma :$ is a composite field of weight 2 whose self-OPE has a quartic pole $(z-w)^{-4}$, but T is *not* a generator: it is determined by β and γ and does not appear in the generating set.

The consequence for the seven-face programme is that the GZ26 commuting Hamiltonians of the $\beta\gamma$ system vanish identically: there is nothing to quantize at the sphere level, because the collision residue $r(z)$ of $\Theta_{\beta\gamma}$ is zero. The shadow depth $r_{\max} = 4$ of $\beta\gamma$ comes not from higher OPE poles but from the quartic contact invariant, which lives on a charged stratum of the cyclic deformation complex and exits the complex by rank-one rigidity (stratum separation).

Remark 4.3 (Stratum separation)*.* Class C is characterized by a mechanism distinct from classes G, L, and M: the termination of the shadow obstruction tower at arity 4 occurs not because the shadow metric Q_L becomes a perfect square (as in G and L) nor because it remains infinite (as in M), but because the quartic contact invariant lives on a stratum

whose self-bracket exits the complex. On the primary line, Q_L is the constant $(2\kappa)^2$, but the full cyclic deformation complex has additional charged strata that carry the contact invariant. This is why class C appears in the classification with the annotation $\Delta = 0^*$: the discriminant vanishes on the primary line but the contact obstruction is nonzero in the full complex.

5 Compute engines

The seven-face programme is backed by fourteen compute engines, each independently verifying one or more face identifications. Every engine has a test file with multiple independent verification paths per tested formula.

Engine	**Tests**	**Faces**
`theorem_gz26_commuting_hamiltonians_engine`	32	F4, F7
`theorem_dnp_meromorphic_tensor_engine`	23	F1, F2
`theorem_bv_brst_genus1_constraints_engine`	59	F3 (genus 1)
`theorem_w3_holographic_datum_engine`	83	F4, F7 ($\mathcal{W}_3$)
`theorem_pva_classical_r_matrix_engine`	61	F3, F5
`theorem_virasoro_spectral_r_matrix_engine`	91	F5 (Vir)
`theorem_nonprincipal_line_operators_engine`	63	F2 (non-principal DS)
`theorem_genus2_planted_forest_gz26_engine`	73	F4 (genus 2)
`theorem_w3_commuting_hamiltonians_engine`	105	F4, F7 ($\mathcal{W}_3$)
`theorem_three_way_r_matrix_engine`	72	F5, F6, F7
`theorem_ainfty_nonformality_class_m_engine`	77	trichotomy (M)
`theorem_dk0_evaluation_bridge_engine`	63	F2, F4
`theorem_sl3_yangian_r_matrix_engine`	88	F5 ($\mathfrak{sl}_3$)
`theorem_three_paper_intersection_engine`	39	invariants, trichotomy
Total	**929**	

The critical engine is `theorem_three_paper_intersection_engine`: its 39 tests check the $(p_{\max}, k_{\max}, r_{\max})$ triple for every row of the three-invariants table and confirm the operator-order assignment. The $\beta\gamma$ row is verified by the dedicated test `test_betagamma_class_C`, which is the sharpest cross-check for the three-invariants distinction: it confirms $p_{\max}(\beta\gamma) = 1$, $k_{\max}(\beta\gamma) = 0$, $r_{\max}(\beta\gamma) = 4$, operator order trivial, class C.

6 Status and loose ends

6.1 Volume status

The seven-face programme is proved for the standard landscape in the three-volume monograph. The per-face per-volume status is:

Face	**Vol I**	**Vol II**	**Vol III**	**Source**
F1 (twisting morphism)	proved	specialized	specialized	Vol I Thm A
F2 (line operator R-twist)	cited	proved	specialized	DNP 2025
F3 (PVA r-matrix)	proved	used	specialized	KZ 2025
F4 (GZ26 commuting)	proved	used	specialized	GZ 2026
F5 (Yangian r-matrix)	proved	cited	specialized	Drinfeld 1985
F6 (Sklyanin bracket)	proved	cited	specialized	STS 1983
F7 (Gaudin generator)	proved	cited	specialized	FFR 1994

6.2 Loose ends

Three loose ends remain in the seven-face programme:

1. *Non-algebraic-family constructions*: the master theorem is proved for algebraic families with rational OPE coefficients. The residual universality conjecture restricts to non-algebraic-family constructions, and no known modular Koszul example with simple Lie symmetry falls outside the algebraic-family class.

2. *Multi-weight genus $g \geq 2$*: for multi-weight algebras at $g \geq 2$, the face identifications receive a cross-channel correction $\delta F_g^{\mathrm{cross}} \neq 0$. The seven-face programme is the uniform-weight-lane projection (Definition **??**); the full multi-weight generalization requires an additional summand from cross-channel stable graphs and is the subject of the multi-weight genus expansion theorem.

3. *Non-principal W-algebras*: arbitrary-nilpotent BV duality (bar-cobar commutes with non-principal DS reduction) is conjectural beyond hook-type in type A. The seven-face programme is proved for principal DS reductions and for the hook-type corridor in type A.

The loose ends do not affect the master theorem on the standard landscape; they precisely record the scope boundaries.

6.3 Anti-patterns to avoid

Three anti-patterns (AP59, AP60, AP61) expose the common failure modes in using the seven-face programme:

- **AP59**: Never conflate the three invariants $p_{\max}$, $k_{\max}$, $r_{\max}$. Writing "shadow depth" when one means "OPE pole order" is wrong in general: the $\beta\gamma$ system has shadow depth 4 and OPE pole order 1.

- **AP60**: Distinguish generator OPE pole order from composite OPE pole order. The Sugawara tensor of $\beta\gamma$ has quartic self-OPE, but the generators β and γ have simple-pole OPE. The bar complex is built from generators, not composites.

- **AP61**: Do not equate shadow depth with operator order without the $k_{\max}$ translation. The GZ26 operator order is $\max(k_{\max} - 1, 0)$, not $r_{\max} - 2$. The two agree for classes G, L, M but disagree for class C.

7 Closing remarks

The seven-face programme is the unifying thread of the three-volume monograph on chiral algebras and modular Koszul duality. Its content is not a new theorem in any single framework: each face $\mathrm{F}i \Leftrightarrow \mathrm{F}j$ identification uses established tools in its source framework. Its content is the *network* of identifications: the assertion that seven different mathematical traditions, working in seven different categorical languages with seven different computational apparatuses, all describe the same underlying element $r(z) \in \mathcal{A}^! \otimes \mathcal{A}^![\![z^{-1}]\!]$ of the Koszul dual.

The surprising aspect is not that equivalences exist: algebraic structures often have multiple presentations. The surprising aspect is that the equivalences close the chain without obstruction, so that starting from any one face and walking the chain returns to the same underlying object. This is the *coherence* of the programme: every face is verified against every other face, both in the mathematics of the monograph and in the 929-test compute layer. Coherence at this scale is not automatic; it is earned, face by face, by checking that the normalization conversions (classical $\hbar \leftrightarrow$ quantum $\hbar$, PVA λ-brackets $\leftrightarrow$ OPE modes, Yangian level shift $\leftrightarrow$ affine level shift) all agree under cross-composition.

The four-class classification G, L, C, M gives the seven faces a structural skeleton. Classes G and L are the "easy" cases: the shadow metric on the primary line is a perfect square, the shadow obstruction tower terminates at arity 3, and the GZ26 Hamiltonians are multiplication operators. Class M is the "generic" case: the shadow obstruction tower is infinite, the Hamiltonians are differential operators of order $2N-2$, and the seven faces all carry nontrivial information. Class C is the "exceptional" case: the shadow obstruction tower terminates at arity 4 by stratum separation, but the collision residue vanishes, so the GZ26 Hamiltonians are trivial. The $\beta\gamma$ system is the archetype of class C, and it is also the critical witness for the independence of the three invariants.

We close with the observation that the seven-face programme is an *audit instrument*: every time a new face (an eighth framework, a new quantization scheme, a new categorical presentation) is added to the network, its identification with the other seven must be proved separately, and the compute engines must be extended to verify it. The programme is not closed: it is open-ended, and each new face is a new testable prediction. The current statement of Theorem 3.1 is the snapshot at seven; future work will extend it to eight or more.

References

[1] T. Dimofte, N. Niu, A. Py, *Line operators in 3d holomorphic-topological gauge theory and categorified Koszul duality*, preprint 2025.

[2] Z. Khan, P. Zeng, *Poisson vertex algebras and 3d sigma models with holomorphic-topological boundary*, preprint 2025.

[3] D. Gaiotto, P. Zeng, *Holographic reconstruction of chiral algebras from commuting sphere Hamiltonians*, preprint 2026.

[4] V. G. Drinfeld, *Hopf algebras and the quantum Yang-Baxter equation*, Dokl. Akad. Nauk SSSR **283** (1985), 1060–1064.

[5] M. A. Semenov-Tian-Shansky, *What is a classical r-matrix?*, Funct. Anal. Appl. **17** (1983), 259–272.

[6] B. Feigin, E. Frenkel, N. Reshetikhin, *Gaudin model, Bethe ansatz and critical level*, Comm. Math. Phys. **166** (1994), 27–62.

[7] R. Lorgat, *Modular Koszul duality: a monograph on bar-cobar duality for chiral algebras on algebraic curves*, Volume I (manuscript, 2026).

[8] R. Lorgat, *A-infinity chiral algebras and* $3d$ *holomorphic-topological QFT*, Volume II (manuscript, 2026).

[9] R. Lorgat, *Calabi–Yau quantum groups and the categorified seven-face programme*, Volume III (manuscript, 2026).